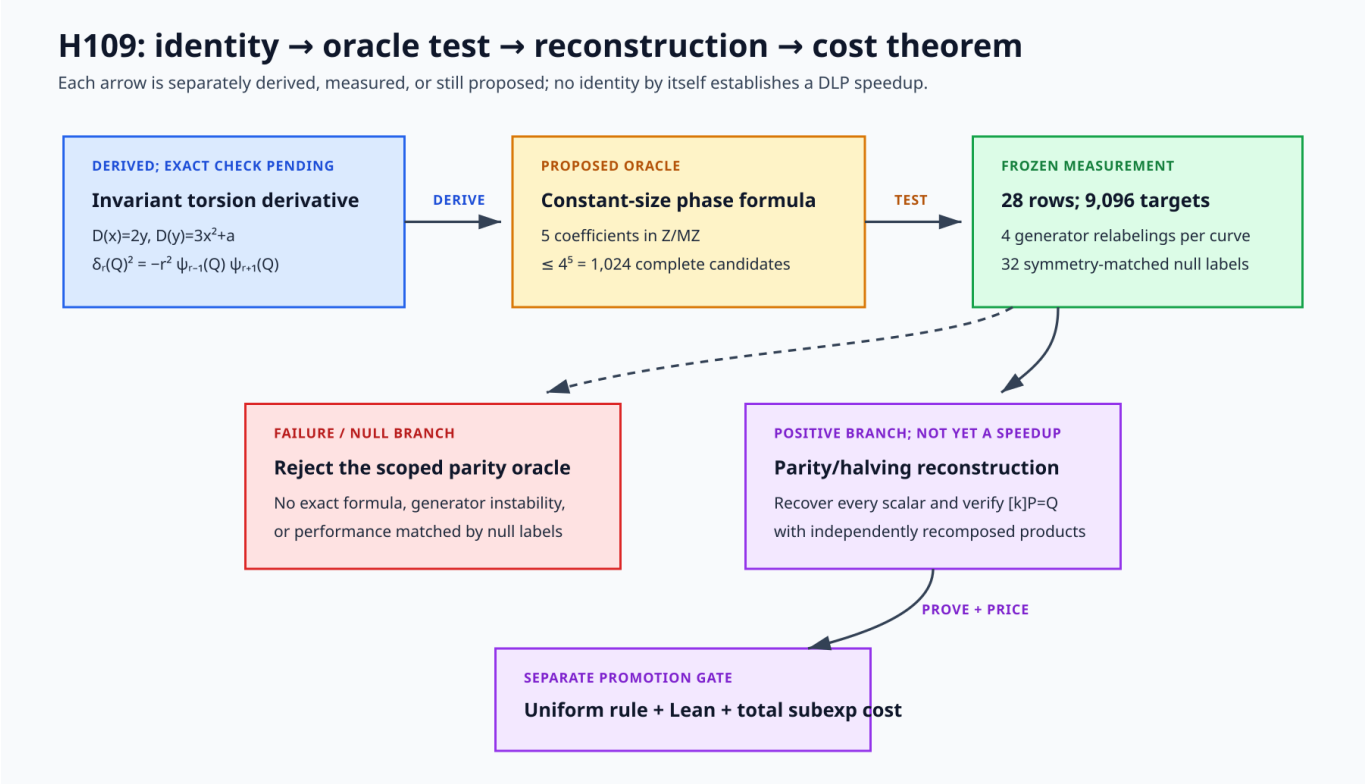
H109 — torsion-local division-polynomial derivative oracle

Date frozen: 2026-10-31



Editable vector source: h109-torsion-derivative-flow.svg.

The diagram separates four claims: the local algebraic identity, existence of an exact compact phase formula, end-to-end scalar recovery, and a uniform subexponential cost theorem. Only the first is mathematically predicted at freeze time; all other arrows are proposed or pending measurement.

Requirement-to-evidence contract

- **Materially new mechanism:** use the invariant local derivative `D psi_r(Q)`, not a fitted span of character codewords.
- **Novelty screen:** preserve database queries, closest sources, and a scoped "novelty not established" conclusion.
- **Measurement:** evaluate every target on seven curves and four generator relabelings; report field, recurrence, character, and decoder costs.
- **Control:** use the paired H107/H108 baseline, local-identity checks, and symmetry-matched random parity labels with the same compact formula budget.
- **Proof:** provide Lean certificates for the derivation, phase/parity interface, and binary reconstruction. A complexity theorem is required for any subexponential claim.
- **PR:** open one branch and one draft PR for H109 before implementation.

Mechanism

For the invariant derivation on a short Weierstrass curve

```
D(x) = 2y,
D(y) = 3x^2 + a,
```

let

```
delta_r(Q) = D(psi_r)(Q),        Q in <P> minus {O}.
```

Every nonzero subgroup point is a simple zero of `psi_r`. Comparing the Laurent expansion of the multiplication formula

x([r]R) = x(R) - psi_(r-1)(R) psi_(r+1)(R) / psi_r(R)^2

with [r]^*(dx/(2y)) = r dx/(2y) predicts the local identity

delta_r(Q)^2 = -r^2 psi_(r-1)(Q) psi_(r+1)(Q).

H109 first checks this identity exactly at every frozen point. It then tests a constant-size family of multiplicative character formulas made from

delta_r(Q), psi_(r-1)(Q), psi_(r+1)(Q), y(Q), constant primitive root.

Unlike H105--H108, the number of fitted coefficients is bounded independently of r. Division-polynomial values and invariant derivatives are evaluated by memoized recurrence with dual-number/product-rule arithmetic.

Frozen formula family and protocol

Let M=2^v2(p-1) and encode maximal base-field character phases in Z/MZ. Completely enumerate every coefficient tuple

(c0,c_delta,c_minus,c_plus,c_y) in (Z/MZ)^5

for the phase of

```
g^c0 delta_r^c_delta psi_(r-1)^c_minus
    psi_(r+1)^c_plus y^c_y.
```

This is at most 4^5=1024 formulas on the frozen curves. No coefficient budget grows with r, and no heuristic search is permitted.

1. Run all seven retained curves, generator multipliers 1,2,3,5, and every nonzero subgroup point.
2. Check psi_r(Q)=0, nonvanishing of the four formula inputs, the predicted squared-derivative identity, and agreement between memoized dual recurrence and a direct small-index symbolic/difference control.
3. Score every frozen formula against canonical scalar parity. Retain exact formulas, best matches, all ties, coefficient vectors, and operation counts.
4. Run 16 symmetry-matched random parity-label controls on the primary p=1 mod 4 row E101 and 16 on the primary p=3 mod 4 row E127. Each control assigns complementary bits to k and r-k and uses the identical complete formula family.
5. Compare with H107/H108's exact but linear-support character modules, charging their setup and support rather than only online character calls.
6. If any exact observed formula exists, execute the parity/halving decoder on every target for that row. A primary positive must replicate across all four generators; the generic gate requires all 28 rows.
7. Independently reevaluate any exact formula through a separately composed field product and validate every reconstructed scalar by EC multiplication.

Runtime and proof gates

The candidate qualifies as subexponential only if all of the following are established:

- a public coefficient rule or the frozen constant-size enumeration constructs an exact formula on every generic row;
- memoized evaluation of the required values and derivatives has a proved poly(log r, log p) or L_r(1/2,O(1)) bound including field operations;
- the parity decoder is exact and its total setup, advice, online work, memory, and verification stay within that bound;
- Lean proves the certificate-to-parity step, reconstruction, and the cost theorem actually used by the claim.

An exact toy formula without the uniform theorem is evidence on the frozen instances, not a generic subexponential algorithm.

Prior-art screen

Queries were run on Crossref, OpenAlex, and arXiv for combinations of “division polynomial derivative,” “torsion point,” “elliptic divisibility sequence derivative,” “theta group local parameter,” and “discrete logarithm.” Closest located sources were:

- Coombes, "Elliptic curves and logarithmic derivatives";
- Silverman, "p-adic properties of division polynomials and elliptic divisibility sequences";
- Burhanuddin and Huang, "Elliptic curve torsion points and division polynomials";
- Lauter and Stange's EDS/ECDLP equivalence.

The bounded search did not locate the frozen derivative-phase parity formula, but it is not source-complete. Status: mechanism-distinct; novelty not established.

The repository lead-atlas screen also found the closest internal predecessor, ECDLP-IDEA-006, and its independent P1540 audit. Those records test low-rank/annihilator and translated-coordinate index location; the audit shows that the QRT state is birationally conjugate to the original elliptic translation and that low-degree rational additive or multiplicative linearization is unavailable. H109 does not claim an annihilator or a new state representation: its separately falsifiable question is whether the *torsion-local invariant derivative* has a constant-size two-power character phase that outputs scalar parity. A positive formula must therefore survive the complete formula enumeration, generator relabelings, matched random-label controls, full scalar reconstruction, and total-cost gate below. This distinction is mechanism-level only and does not establish novelty.

PR gate

Implementation may start only after this freeze is present in a draft H109 PR. The canonical experiment PR belongs in `isogeny-labs/crypto`; the paired reference-harness PR belongs in `isogeny-labs/cryptanalysis`. Neither PR is a result until the frozen matrix has run and its artifacts are committed.